BINF 6110 Assignment 1:

Demonstration of *S. Enterica* Genome Assembly, Annotation, and Alignment.

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Background:

Genome assembly is a vital process to modern genetic analysis pipelines. For most forms of genetic sequencing, a large number of DNA reads (nucleotide sequences) of unclear origin and variable length are produced. In order to make use of this data, genome assembly is required to sort these reads, determine their origin, and map these reads in relation to each other. This is especially important for use in aligning reads to other genomes, where proper sequence and orientation can be vital for determining identity and finding differences. Depending on the species, sample source, sequencing technology, and intended use, the best genome assembly tool for a given analysis can vary significantly. When the subject of concern is pathogenic, this is emphasized all the further. (Bogaerts et al., 2025)

The main differences in genetic analysis approaches begin with the type of sequencing technology used to produce the reads. With the exception of short RNA sequences and viroids, genomes and DNA/RNA strands were typically too long to accurately analyze as a single molecule in a practical timeframe until the refinement of long-read sequencing. As such, whole genome sequencing had been performed using fragment based short-read approaches like Illumina sequencing. In such sequencing, the tested genome is intentionally cut into short sequences of a couple of hundred nucleotides in length (typically, 150-200nt, though wider ranges exist) to benefit from parallel sequencing. Overlapping motifs between fragments are used to reconstruct a model of the original genome via assembly.

This short read sequencing, while accurate, has a problem assembling the fragments into a larger architecture, or dealing with repeated sequences. Contiguous reads (contigs), while allowing fragments to be put together in larger groups, may not be able to overlap into a single unified whole, forcing assembly software to guess at cross-contig order. If motifs within a read are repeated for longer than the length of that read, it may not be possible to accurately

tell how many of the reads are due to duplicate fragments as an artifact of sequencing and how many are due to genuine repeats in the original genome.

This challenge has knock-on effects for later alignment to a reference genome, which is vital for comparison between samples. To be useful for comparative analysis, a sample sequence must be aligned against a known sample, and differences compared. Without an accurate repeat count or genomic architecture, major sources of variation between individuals and species can be overlooked, misattributed, or mis-identified. For example, a translocation event may be missed entirely if the relevant gene was only found in an isolated contig without sufficient surrounding context sequenced.

Long read sequencing, on the other hand, attempts to preserve DNA/RNA strands intact and sequence them as one unit, using technological advances such as electric-resistance-reporting synthetic membrane-bound proteins (Oxford Nanopore) to sequence nucleotide strands far faster than traditional methods. As a consequence of the long read length, overlap sequences between strands are far more likely, allowing the sequence to be assembled into more complete, less fragmented contigs.

Earlier in their development, however, this increased architectural consistency came at a cost in individual nucleotide accuracy. The longer a single read, the higher the chance of accumulating error from a variety of factors (In recent years, however, long read sequencing such as Oxford Nanopore has grown sufficiently accurate to be comparable to short read sequencing while addressing these architecture issues.

This paper aims to outline common goals and challenges of assembling a genome and aligning it to a reference genome. Then, using that information, the script aims to take a case study in the form of FastQ reads from an Oxford Nanopore R10 sequence for a *Salmonella enterica* sample (Bogaerts et al., 2025), as an example of both interesting genomics and pathogenic importance.

These reads are to be assembled into a consensus genome, compared to a reference genome, and have any variants visualized as both an assembly and a functional; annotation alignment with multiple useful metrics of comparison to highlight variants of interest in the

sequenced sample. The code to do so is assembled in a functional demonstration vignette as an executable bash script in the attached github repository, replicated below for sake of completeness:
https://github.com/WFeinman/BINF_6110_Assignment_1_S_enterica_assembly_and_alignment

Methods:

Sequence quality was confirmed by running FastQC on the input fastq file. Though specialized for short reads, FastQC is a way of flagging pressing issues in a dataset, such as a low quality score or bad reads. Here, it identified done. Genome assembly was performed using flye version 2.9.6-b1802. Flye is a longstanding, well documented genome annotation software with simple installation and quick operation, making it highly beneficial as a learning tool. Additionally, it is specialized as a long-read assembly tool with explicit argument support for R10 Oxford Nanopore reads such as ours. Other assembly software may be more accurate, but take longer and are more technically involved. Unicycle in particular, though a recent paper showed it had very promising performance metrics, has two other competing assembly software programs put out by the same developer, and is in maintenance rather than active development.

Genome alignment was performed using minimap2 version 2.28-r1209. File conversion with samtools version 1.23.1. Subsequent genome annotation was performed with Clair3 v2.0.0 on the un-assembled alignment, as it has beenone of the top performing annotation packages in recent years (Helal et al., 2022). Finally .bam files were viewed with Integrative Genomics Viewer for Windows Desktop version 2.19.7, comparing to original reference genome from GenBank for S. Enterica.

S. enterica genomic data and Genbank reference genome download locations included in script file and github readme. All operations performed on Windows Subsystem for Linux version 2.5.10.0 bash shell, save for final viewing on the Integrative Genomics Viewer (IGV) for Windows version 2.19.7.

Notably, the long-read dataset has been pre-selected for high read quality and no exceptionally long reads (expected accuracy score Q20+, N50:5-15kb), mitigating potential assembly inaccuracies and falling well within the working range of our alignment software.

If working with lower expected accuracy such as from an earlier Oxford-nanopore set, the assembly portion of the script should be adjusted accordingly, such as by using the "–nano-raw" argument for the flye command in place of the "–nano-hq" to indicate a lower expected accuracy. If there were a few reads in particular of lower quality than the rest, they could be filtered out with a quality score rule to preserve the rest of the data.

Bash Script Execution:

Ensure flye, FastQC, minimap2, and samtools are properly installed and active. Place the "s_enterica_flye_to_minimap_to_bam.sh" script in desired file location.

Give the script execution privileges with: chmod +x s_enterica_flye_to_minimap_to_bam.sh

- And for repeated scripts:
  chmod +x s_enterica_no_fastq_download_flye_to_minimap_to_bam.sh

Execute the script with: bash s_enterica_flye_to_minimap_to_bam.sh

On execution, requisite genomic data and reference genome will be downloaded to the working directory. Alternatively, should the file SRR32410565.fastq already be downloaded from a previous script, an optional script without that specific download has been included for ease of repetition:

bash s_enterica_no_fastq_download_flye_to_minimap_to_bam.sh

Either way, the above genome assembly and alignment will be conducted on the genomic data using the downloaded reference.

Once completed, a "test1.sorted.bam" file will be generated in the working directory. This may be loaded into the IGV and compared to the reference genome

"GCF_000006945.2_ASM694v2_genomic.fna" for the assembly. The "test2.sorted.bam" file may be likewise compared to the reference for the aligned annotation.

Primary Script file contents:

```
#!/bin/bash

#SRA toolkit required for requested file interpretations, install instructions https://github.com/ncbi/sra-tools/wiki/02.-Installing-SRA-Toolkit

#Example flye package installation with bioconda if required. If a non-conda installation is desired, please see https://github.com/mikolmogorov/Flye/blob/flye/docs/INSTALL.md for alternatives.

#conda env create -n flye

#conda activate flye

#conda install flye


#minimap installed via compiling binary executable directly. See https://github.com/lh3/minimap2 for first time installation or alternatives.

#samtools installed directly with "$sudo apt install samtools". See https://www.htslib.org/download/ for alternatives

#clair3 installed via singularity container: $singularity pull docker://hkubal/clair3:latest. See https://github.com/HKU-BAL/Clair3?tab=readme-ov-file#installation for alternatives

#download locations for S.enterica data:

# S.enterica SRR file:https://sra-pub-run-odp.s3.amazonaws.com/sra/SRR32410565/SRR32410565

# S.enterica reference genome: https://ftp.ncbi.nlm.nih.gov/genomes/all/GCA/000/006/945/GCA_000006945.2_ASM694v2/GCA_000006945.2_ASM694v2_genomic.fna.gz

# S.enterica annotation: https://ftp.ncbi.nlm.nih.gov/genomes/all/GCA/000/006/945/GCA_000006945.2_ASM694v2/GCA_000006945.2_ASM694v2_genomic.gtf.gz


#Actual data downloads and unpacking

wget -nc https://sra-pub-run-odp.s3.amazonaws.com/sra/SRR32410565/SRR32410565

#works, just takes it's sweet time. Like 10-20 minutes.

fasterq-dump SRR32410565

seqtk fqchk SRR32410565.fastq | head
```

```
wget -nc
https://ftp.ncbi.nlm.nih.gov/genomes/all/GCA/000/006/945/GCA_000006945.2_ASM694v2/GCA_000006945.2_ASM694v2_genomic.fna.gz


wget -nc
https://ftp.ncbi.nlm.nih.gov/genomes/all/GCA/000/006/945/GCA_000006945.2_ASM694v2/GCA_000006945.2_ASM694v2_genomic.gtf.gz

gunzip GCA*.gz

#Genome assembly with flye:

flye --nano-hq SRR32410565.fastq -o Flye_Assembly -t 12

seqtk fqchk Flye_Assembly/assembly.fasta | head

#Genome alignment with minimap. -t12 used due to a 12-core machine for multithreading, substitute as desired.:

minimap2 -ax map-ont -t12 \

   GCA_000006945.2_ASM694v2_genomic.fna \

   Flye_Assembly/assembly.fasta > test1.sam

#File conversion with samtools for

samtools view -bS test1.sam -o test1.bam

samtools sort test1.bam -o test1.sorted.bam

samtools index test1.sorted.bam

samtools faidx GCA_000006945.2_ASM694v2_genomic.fna

minimap2 -ax map-ont -t 12 \

  GCA_000006945.2_ASM694v2_genomic.fna \

  SRR32410565.fastq | \

  samtools sort -o test2.sorted.bam && \

  samtools index test2.sorted.bam


#Variant calling with clair3

singularity pull docker://hkubal/clair3:latest


mkdir clair3_out

#note: absolute path required for singularity variant calling, so exporting current pwd for later retrieval

ABSOLUTE_PATH=$(pwd)

export STORED_FILE_PATH=$ABSOLUTE_PATH
```

```
singularity exec \
 -B $(echo $STORED_FILE_PATH),$(echo $STORED_FILE_PATH)/clair3_out\
 clair3_latest.sif \
 /opt/bin/run_clair3.sh \
 --bam_fn=$(echo $STORED_FILE_PATH)/test2.sorted.bam \
 --ref_fn=$(echo $STORED_FILE_PATH)/GCA_000006945.2_ASM694v2_genomic.fna \
 --threads=12 \
 --platform="ont" \
 --model_path="/opt/models/r1041_e82_400bps_sup_v500" \
 --include_all_ctgs \
 --output=$(echo $STORED_FILE_PATH)/clair3_out \
 --no_phasing_for_fa \
 --include_all_ctgs \
 --haploid_precise \
 --enable_variant_calling_at_sequence_head_and_tail
gunzip clair3_out/merge_output.vcf.gzm
mv clair3_out/merge_output.vcf merge_output.vcf
```

Results:

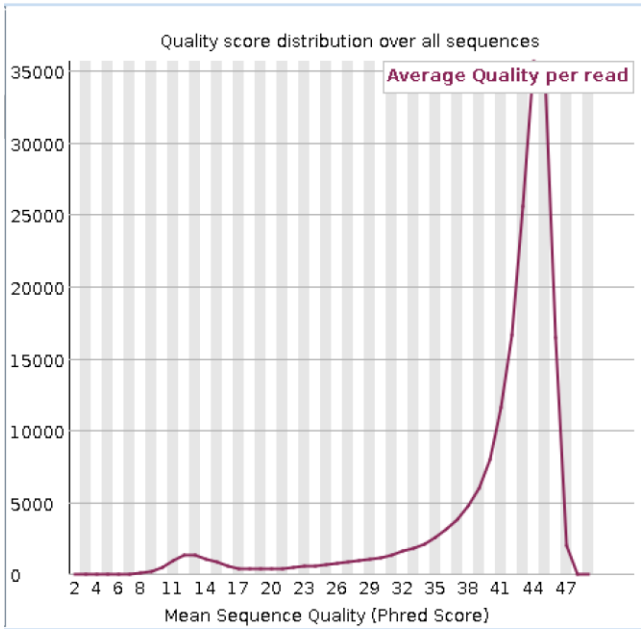

Figure 1: Read quality was confirmed with FastQC. Phred score well above 20 indicates accurate reads.

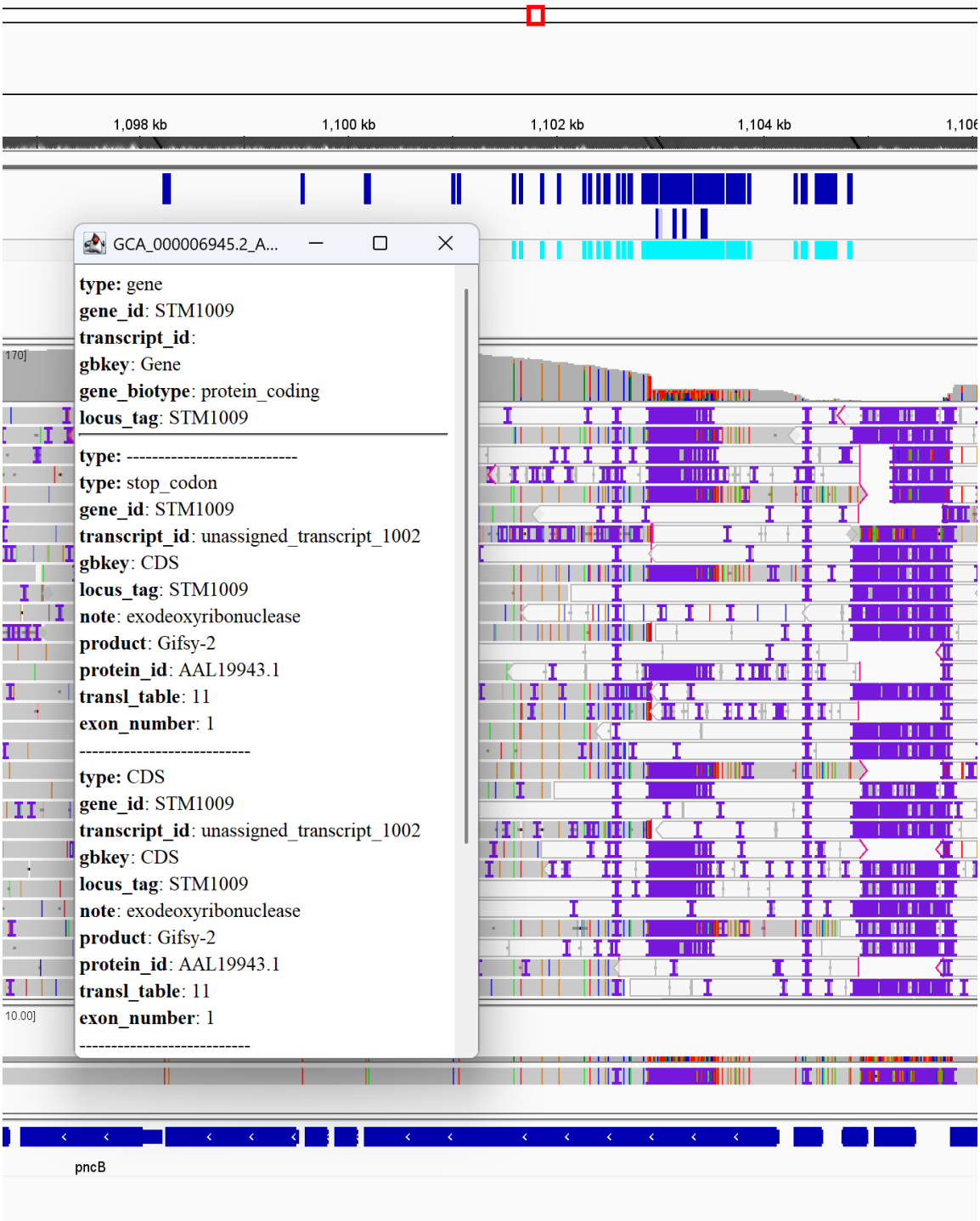

Figure 2: Example cross-file comparison chart. Top-mid histogram thickness indicates extent of coverage in region. Color change on same indicates variance from reference. Red, green, blue, or orange indicates T, A, C, G substitution (respectively) in place of the reference nucleotide. Purple indicates insertion. White indicates a read gap. Example of a low coverage region being highly variable from reference.

Figure 3: Example of distinct chimeric alignment. Read does not map cleanly to aligned region, so supplementary alignment recorded at the closest match site.In this case, a high coverage regtion shows high variation from the reference.

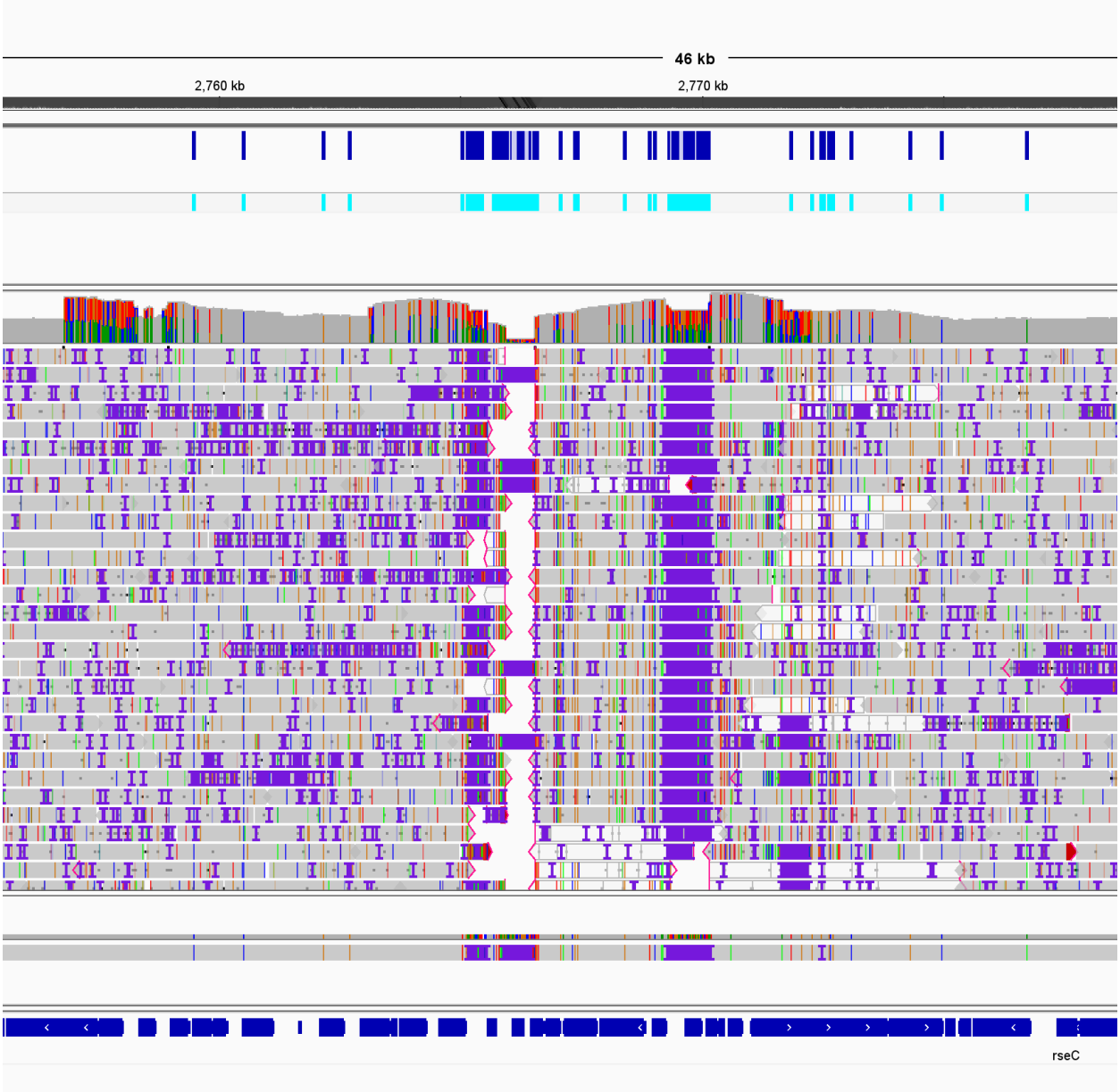

Figure 4: Example of secondary alignment. Unlike chimeric alignments, reads clearly map to aligned reference region, but a secondary motif is actually more common than the reference sequnce at a given spot. In this case, a conserved insertion sequence

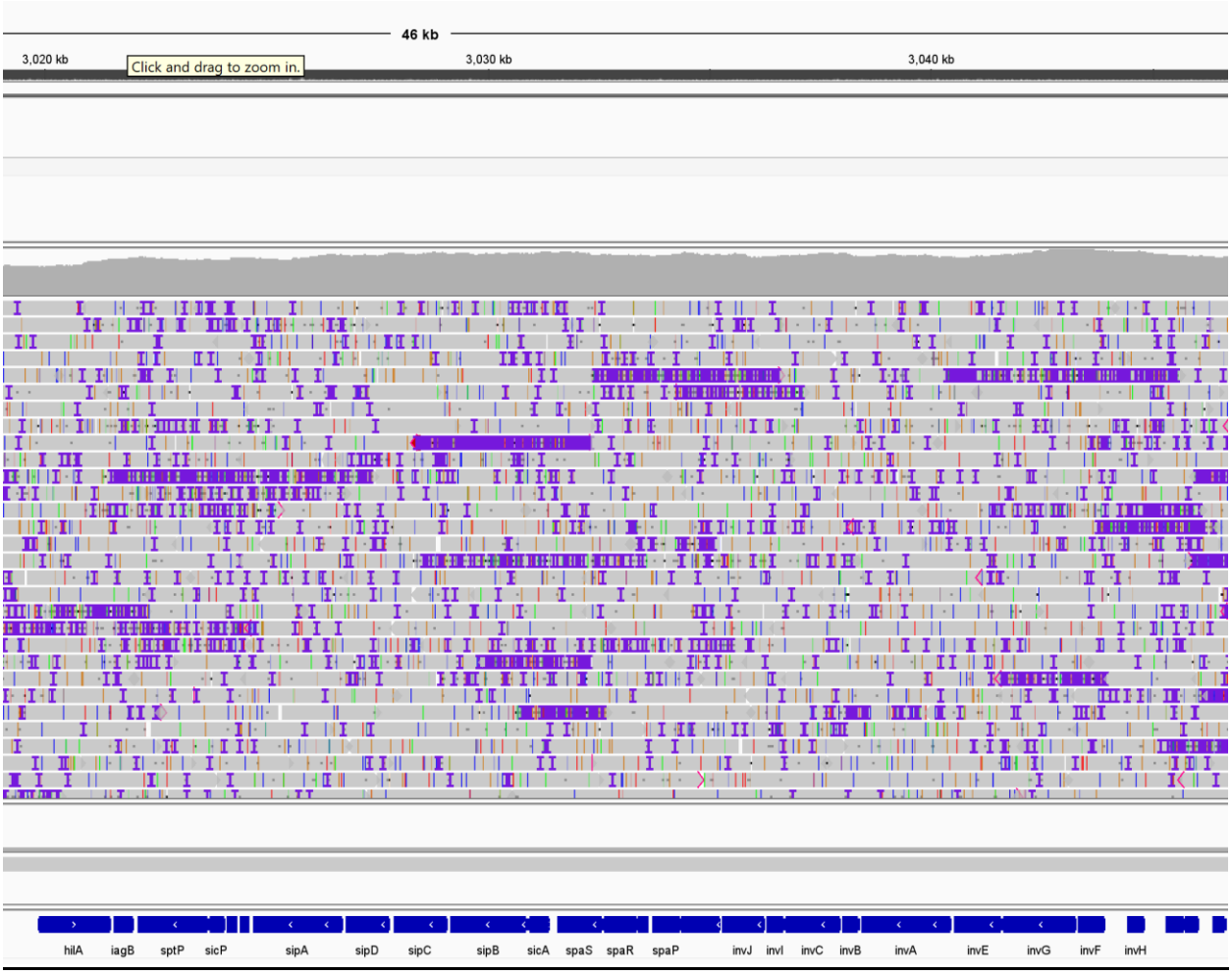

Figure 5: Example of a Salmonella pathogenecity island (SPI-1). Despite an absence of clear variant calls like prior examples, many reads show increased insertions in this broad area.

Discussion:

Overall, the *S. enterica* daataset showed a remarkable number of illustrative alignment features, serving as a good illustrative example of some of the challenges in categorizing genomes even with and accurate dataset. (SAM 2025). Even with high quality long reads, low coverage of a region can lead to interpretation difficulty. Spotty coverage may lead to important regions being under-identified. Disruption from chimeric or secondary alignments can allow important traits to be mis-identified. *Salmonella* species are a highly illustrative example of this. (Sia et al., 2025)

Such species are economically devastating, highly destructive to animal species, and have proven resilient to medical intervention (Jajere, 2019). Plasmid swapping, co-infection, and mutable genomes provide substantially increase variability and evolutionary flexibility, novel functions being gained and lost through both conventional and unconventional methods of gene transfer. (Tautz & Domazet-Lošo, 2011)

Salmonella's pathogenicity islands (SPIs) are an illustration of this behaviour in action. These are genetic regions shared across *Salmonella* species encoding traits which increase bacterial virulence and harm to their host. SPI-1, for example, designates a collection of genes which code for an injectisome protein complex (SipB, SipC, SipD) aiding in host cell invasion. Other SPIs might affix more firmly in host guts, evade immune response, increase toxicity, or a variety of other deleterious effects. (Lou et al., 2019) (Wood et al., 2025)

These regions are not always, however, easily identifiable and trackable. As with other parts of the *Salmonella* genome, these motifs may duplicate, split, translocate, and otherwise be modified and move wildly across genomes and species. These genetic effects provide challenges to traditional bioinformatic analysis, stressing the importance of improving existing tools and developing new ones. Assembly may struggle when gene location and content is highly variable even within the same species. Alignment is difficult in such cases even with deep, accurate sequencing, as the genetic relations they are trying to capture are often non linear, such as horizontal gene transfer.

It should be emphasize that what is captured here was a particularly deep, accurate sequencing dataset and extensive coverage, which lets us characterize the gene interactions here with much more confidence. In studies lacking the time or funding necessary to obtain that accuracy and depth, capturing these shifts becomes far more difficult. The importance of a robust genetic analysis pipeline is emphasized, expecially in light of the last pandemic.

References:

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- Direct Download: https://sra-pub-run-odp.s3.amazonaws.com/sra/SRR32410565/SRR32410565

- Reference Genome: https://ftp.ncbi.nlm.nih.gov/genomes/all/GCA/000/006/945/GCA_000006945.2_ASM694v2/GCA_000006945.2_ASM694v2_genomic.fna.gz

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